

Zikai Zhao

☎ +86-18833116940 | @ zikai3@illinois.edu

EDUCATION

Zhejiang University

B.Eng., GPA: 3.88/4.0, Rank: 4/71

Major: Electrical Engineering and Automation (Dual Degree)

Zhejiang, China

Sep 2023 – Jun 2027

University of Illinois Urbana-Champaign

B.S., GPA: 3.96/4.0

Major: Electrical Engineering (Dual Degree)

Urbana, IL, USA

Sep 2023 – Jun 2027

PUBLICATIONS

Zikai Zhao. “A New Design of 4-Bit Absolute Value Detector.” *Proceedings of the 3rd International Conference on Computer Science and Mechatronics (ICCSM 2025)*, SciTePress, 2025, pp. 122–127. DOI: 10.5220/0014000300004918.

RESEARCH EXPERIENCE

4-Bit Absolute Value Detector Design and Optimization

Sept 2024 – Dec 2024

Supervisor: Prof. Dejan Markovic, UCLA

- Design a 4-Bit Absolute Value Detector by deriving a minimum Boolean expression through a 128-entry truth table
- Perform critical path analysis and gate sizing optimization, and reduce the delay to 33.7 unit delay
- Constrains the delay to $1.5 \times$ the minimum delay and reduces energy consumption to 60% of the original design.

Millimeter-Wave Radar Based Non-Destructive Fruit Sugar Measurement

Jun 2024 – Aug 2024

Supervisor: Prof. Yong WANG, Zhejiang University

- Applied millimeter-wave radar reflections and mathematical processing to estimate fruit sugar content using multiple machine learning models.
- Conducted sweetness measurements, refined the testing interface UI, and introduced repeated computations to improve stability.
- Implemented a non-destructive method for classifying apples into multiple sweetness levels without physical damage.

ACADEMIC PROJECT EXPERIENCE

FPGA Game Development: *Tank Turmoil*

Nov 2025 – Dec 2025

Supervisor: Prof. Zuofu Chen, UIUC

- Programmed FPGA using SystemVerilog to implement a dual-player tank battle game featuring movement, shooting, item collection, and interactive audio effects.
- Designed hardware logic for tank rotation, wall collision detection, bullet generation, and VGA pixel output.
- Handled GPIO keycode inputs to control movement, firing, and rotation, ensuring smooth gameplay on randomly generated maps.

Unreal Engine 5 FPS Game Development

Oct 2025 – Dec 2025

Supervisor: Prof. Eric Shaffer, UIUC

- Designed and implemented core gameplay systems, including health bars, experience and leveling mechanics, point-allocation UI, in-game shop, and purchasing interfaces.
- Developed boss AI behaviors and attack patterns, and managed multi-level scene transitions with persistent player state.
- Coordinated team activities, organized regular meetings, and tracked project progress.

COMPETITIONS

The 6th Zhejiang University Concrete Dragon Boat Competition

May 2025

Supervisor: Prof. Yushi Cheng, ZJU

- Led the design of a custom concrete boat under strict dimensional constraints, balancing trade-offs among speed, maneuverability, and structural integrity
- Awarded Third Prize at the campus level

HONORS & AWARDS

Dean's list, UIUC,

Fall 2025

Third Prize, Concrete Dragon Boat Competition, Zhejiang University

May 2025

University-level Academic Excellence Award, Zhejiang University, Top 15%

Dec 2024

University-level Third-class Scholarship, Zhejiang University, Top 15%

Dec 2024

ADDITIONAL INFORMATION

Programming Languages & Tools: C, C++, Python(Numpy, pandas, signal, pyplot, scikit-learn), SystemVerilog, LC-3, Linux